

STUDENT ACTIVITY HANDOUT

IFY STATISTICS | SEMESTER 1 - WEEK 1B

Mathematical Foundations II: Algebra, Units, Percentages and Ratios

Bellerbys
Preparation for University Education

Name _____ Class _____ Date _____

EXERCISE 1

Label the Mathematics

Use the equation below. Circle the variable, underline the coefficients and box the constants. Then identify the terms, like terms, left side and right side.

$$7y - 3y + 4 = 20$$

NOTES AND LABELS

EXERCISE 2

Balance Chain

Solve $5x - 3 = 2x + 9$. Apply the same operation to both sides, isolate x , then check your answer in the original equation.

CALCULATIONS

BRACKETS AND LIKE TERMS

EXERCISE 3

Brackets and Like Terms

Solve each equation. Show enough working for another student to follow.

1. Solve $5y + 2(y - 1) = 3y + 10$.

CALCULATIONS

2. Challenge: solve $(2z - 1)/5 - z = (4 - 3z)/4$.

Hint: multiply every term by 20 first to remove the fractions.

CALCULATIONS

Rewrite each equation with zero on one side, then decide whether it is quadratic.

EQUATION	QUADRATIC? IF YES, WRITE STANDARD FORM
1. $2x + 5 = 0$	
2. $x^2 + 5x + 6 = 0$	
3. $x^3 - 2 = 0$	
4. $4x^2 = 9$	

Solve $2x^2 + 5x - 3 = 0$. Identify a, b and c, then use the formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

a = _____

b = _____

C = _____

CALCULATIONS

UNITS AND CONVERSIONS

EXERCISE 6

Convert and Check

Show the calculation and include the final unit.

1. Convert 750 mL to litres.

CALCULATIONS

2. Convert 12 km to miles. Use 1 mile $\approx$ 1.609 km. Give the result to 3 d.p.

CALCULATIONS

3. Convert 100 inches to centimetres. Use 1 inch = 2.54 cm.

CALCULATIONS

EXERCISE 7

Conversion Safety Decision

A 2,400 kg packaging machine must be lifted. The forklift capacity is 5,000 lb. Use 1 kg $\approx$ 2.20 lb. Convert the mass and decide whether the lift is safe. State the evidence.

CALCULATIONS

PERCENTAGES**EXERCISE 8****Choose the Percentage Method**

Calculate each answer and show your method.

1. Find 35% of 240.

CALCULATIONS

2. 18 is what percentage of 72?

CALCULATIONS

3. 42 is 35% of what number?

CALCULATIONS

4. A £280 item has 15% off. Find the sale price.

CALCULATIONS

EXERCISE 9**Starting-Value Trap**

A product falls from £900 to £700. Find the change and percentage decrease to 1 d.p. Explain why dividing by £700 is wrong.

CALCULATIONS

RATIOS AND PROPORTIONS**EXERCISE 10****Ratio Interpretation Lab**

Read the wording carefully, then calculate.

1. Simplify 45:60.

CALCULATIONS

2. $A:B = 3:2$. If $A = 18$ L, find B .

CALCULATIONS

3. Divide 70 students into two groups in the ratio 2:5.

CALCULATIONS

4. For the two-part ratio 4:5, find the first quantity as a percentage of the second, then as a percentage of the complete total. Give the second result to 1 d.p.

CALCULATIONS

EXERCISE 11**Missing-Total Proportion**

9 out of every 10 customers recommend a product. If 450 customers recommended it, find the total surveyed, the number who did not recommend it, and the scale factor.

CALCULATIONS

BRAIN-BOILING CHALLENGE**EXERCISE 12** **Brain-Boiling Challenge**

Show every step and check whether your final answers make sense.

1. Solve $2x^2 + 5x - 12 = 0$ by factoring and by the quadratic formula.

Factoring hint: split $+5x$ into $+8x - 3x$.

FACTORING

QUADRATIC FORMULA

2. A machine has mass 3,000 kg. A forklift has a 5,000 lb limit. Use $1 \text{ kg} \approx 2.20 \text{ lb}$. Is the lift safe, and what percentage above the limit is the load?

CALCULATIONS

INDEPENDENT EXIT TICKET

EXERCISE 13 Independent Exit Ticket

Work independently. Show your calculations and check each answer.

1. Solve $3x + 5 = 20$.

CALCULATIONS

2. Solve $x^2 - 5x + 6 = 0$ by factoring.

CALCULATIONS

3. Convert 3 miles to kilometres. Use 1 mile $\approx$ 1.609 km. Give the result to 3 d.p.

CALCULATIONS

4. Find the percentage increase from £40 to £50.

CALCULATIONS

5. Simplify the ratio 16:24.

CALCULATIONS
